

FACTORYLENS AI

See the defect.
Find the reason.
Understand the impact.

Research & Prototype Concept for AI-Assisted
Manufacturing Decision Support

QUALITY → PRODUCTION → ECONOMICS

Prepared from the provided hackathon problem statement and reference images.
Reference visuals are illustrative; organizer data is not yet available.



1. The Manufacturing Environment

BACKGROUND

A factory is a connected system—not a collection of isolated machines.



What happens in a typical operation?

Materials and products move through multiple stages.

Each stage affects quality, speed, capacity and cost.

A local problem can create a wider production impact.

Decision-makers need one view that connects these signals.

Core research idea: connect signals across the factory.

2. The Problem: Quality Is Not the Whole Story

PROBLEM

Detecting a defective unit is useful—but decision support must go further.

From inspection to understanding

- | | | |
|----|------------------|--|
| 01 | Detect | Is the unit acceptable or defective? |
| 02 | Localize | Where is the defect, when supported by data? |
| 03 | Correlate | Which batch/process patterns are associated? |
| 04 | Impact | What could this mean for flow and economics? |

The goal is not just a classifier. It is a traceable decision-support system.



3. Quality Inspection & Defect Detection

QUALITY

Computer vision can support consistent inspection at scale.



What the system should communicate

Accept / defective classification

Defect type when supported by training data

Localization using boxes, masks or heatmaps when available

Confidence score instead of overconfident decisions

A review path for uncertain or novel cases

Human-in-the-loop: AI supports decisions; it does not replace judgment.

4. Why False Rejects & Uncertainty Matter

ROBUSTNESS

A useful inspection system must know when it is unsure.

Two types of risk

FALSE

Rejecting an acceptable unit

MISS

Failing to flag a real defect

Research direction

Show confidence, not only a label.

Allow uncertain cases to be reviewed.

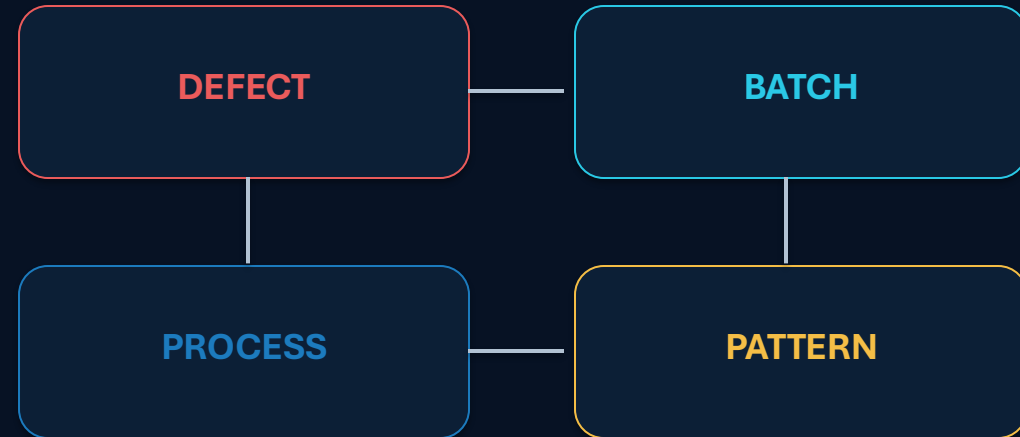
Separate known defect classes from possible novel defects.

Track false-reject patterns during evaluation.



5. Root-Cause Correlation: From “What?” to “Why?”

The system should connect inspection results with production context.



Evidence-based interpretation

Compare defect rates across batches or process conditions.

Highlight possible contributing factors.

Show the evidence behind each relationship.

Avoid claiming causation unless the data supports it.

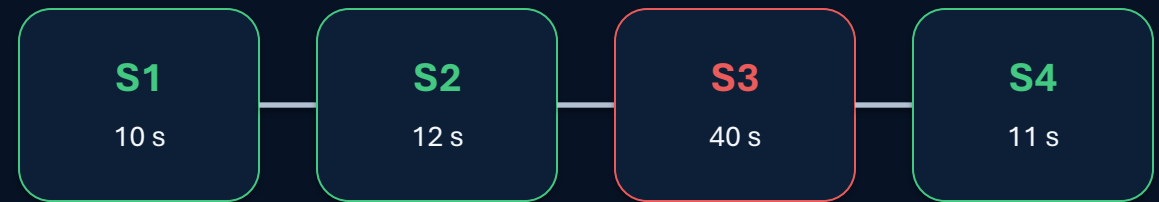
6. Production Bottleneck Analysis

PRODUCTION

A bottleneck is a stage whose limited capacity or high processing time restricts overall flow.



Example production flow



S3 = bottleneck

Detect unusually slow stages.

Measure queue / waiting patterns where data allows.

Estimate throughput loss linked to the bottleneck.

Connect bottlenecks with quality and downtime signals.

7. From Production Loss to Economic Impact

A defect or delay matters because it can consume time, capacity and money.

Defect



Rework / Scrap

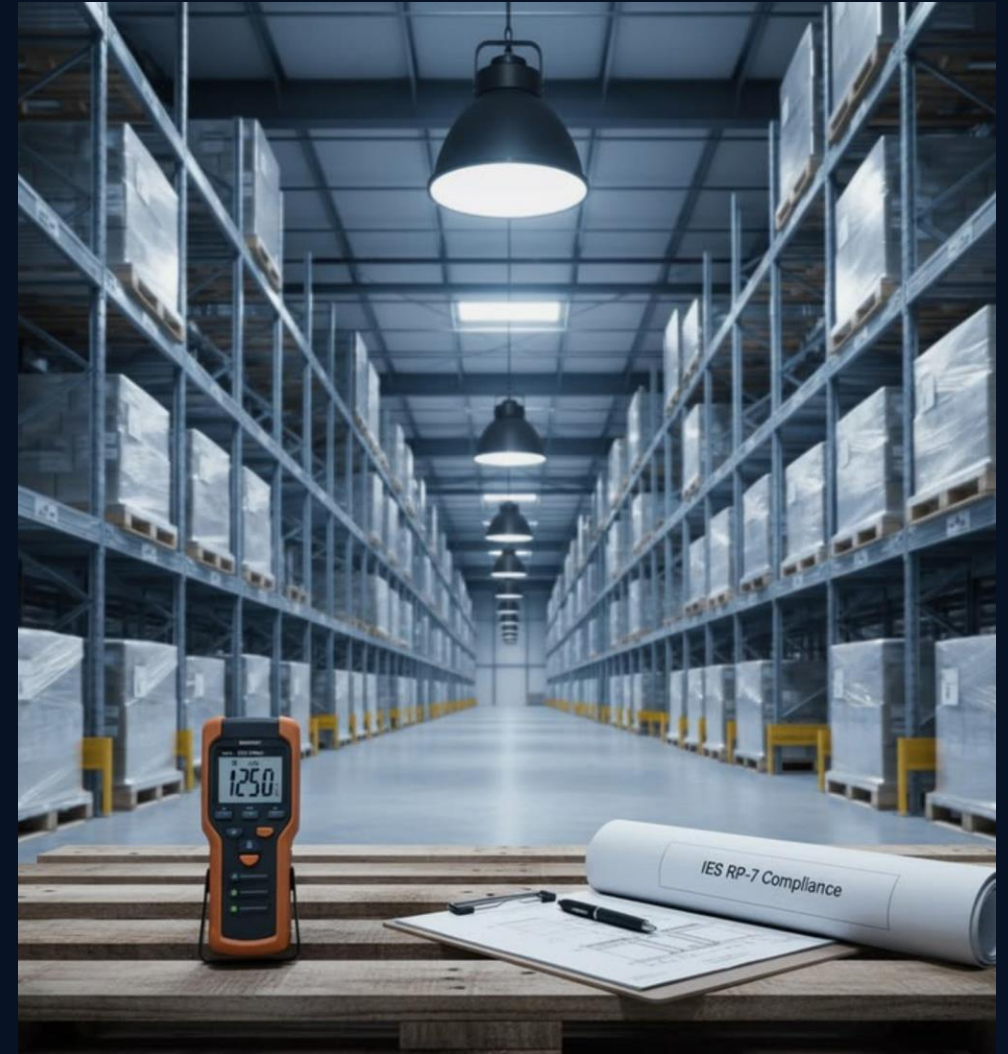


Capacity Loss



Potential Cost Impact

Important: economic values should come from organizer data or clearly labeled assumptions.



8. The Innovation: The Problem Chain

INNOVATION

One traceable chain connects quality, production and economics.

DEFECT

WHERE / WHEN

PROCESS PATTERN

BOTTLENECK

THROUGHPUT IMPACT

ECONOMIC IMPACT

RECOMMENDATION



This is the core differentiation: not just “find the defect,” but “show the connected impact.”

9. Proposed System Architecture

ARCHITECTURE

Software-only decision support with clear boundaries.



INPUT DATA

Inspection images • Production records • Batch/process data • Economic data

AI / ANALYTICS

Defect model • Correlation • Bottleneck detection • Impact estimation

DECISION LAYER

Evidence • Confidence • Uncertainty • Recommended actions

USER INTERFACE

Dashboard • Defect view • Problem Chain • Bottleneck • Recommendations

No direct machine control • Advisory / simulated interventions only

10. Prototype Dashboard: What the Judge Should

PROTOTYPE

See

A simple flow makes the research idea easy to demonstrate.

FactoryLens AI / DEMO — SIMULATED DATA

8.4%

Defect rate

S3

Bottleneck stage

2.7 h

Estimated downtime

₹1.2L

Potential impact*

DEFECT ANALYSIS

Defect type: Surface anomaly

Confidence: 91%

Review status: Confirmed

PROBLEM CHAIN

**Defect → Batch 17 → Station 3
→ Throughput risk → Cost impact**

Evidence + confidence shown

ADVISORY

**Investigate Station 3 process
conditions and Batch 17
correlation.**

Not an automatic machine command.

*Illustrative value only; replace with organizer data.

11. Research & Implementation Roadmap

Build the smallest credible version first, then deepen the analytics.

PHASE 1	Understand & map	Problem analysis • data fields • success criteria
PHASE 2	Prototype	FlutterFlow dashboard • simulated data • Problem Chain
PHASE 3	AI baseline	Defect classification/localization using available dataset
PHASE 4	Analytics	Correlation • bottleneck • throughput • economic impact
PHASE 5	Validation	False rejects • uncertainty • explainability • demo testing

Current status: no organizer dataset yet → use clearly labelled simulated data until the real dataset arrives.

12. What Makes the Concept Different?

DIFFERENTIATION

The differentiator is the connection between three business layers.

QUALITY

What went wrong?

PRODUCTION

Where is flow affected?

ECONOMICS

What could be the impact?

Instead of three disconnected dashboards:

FactoryLens AI creates one evidence-based story.

DEFECT → PROCESS PATTERN → BOTTLENECK → IMPACT →
ADVISORY ACTION



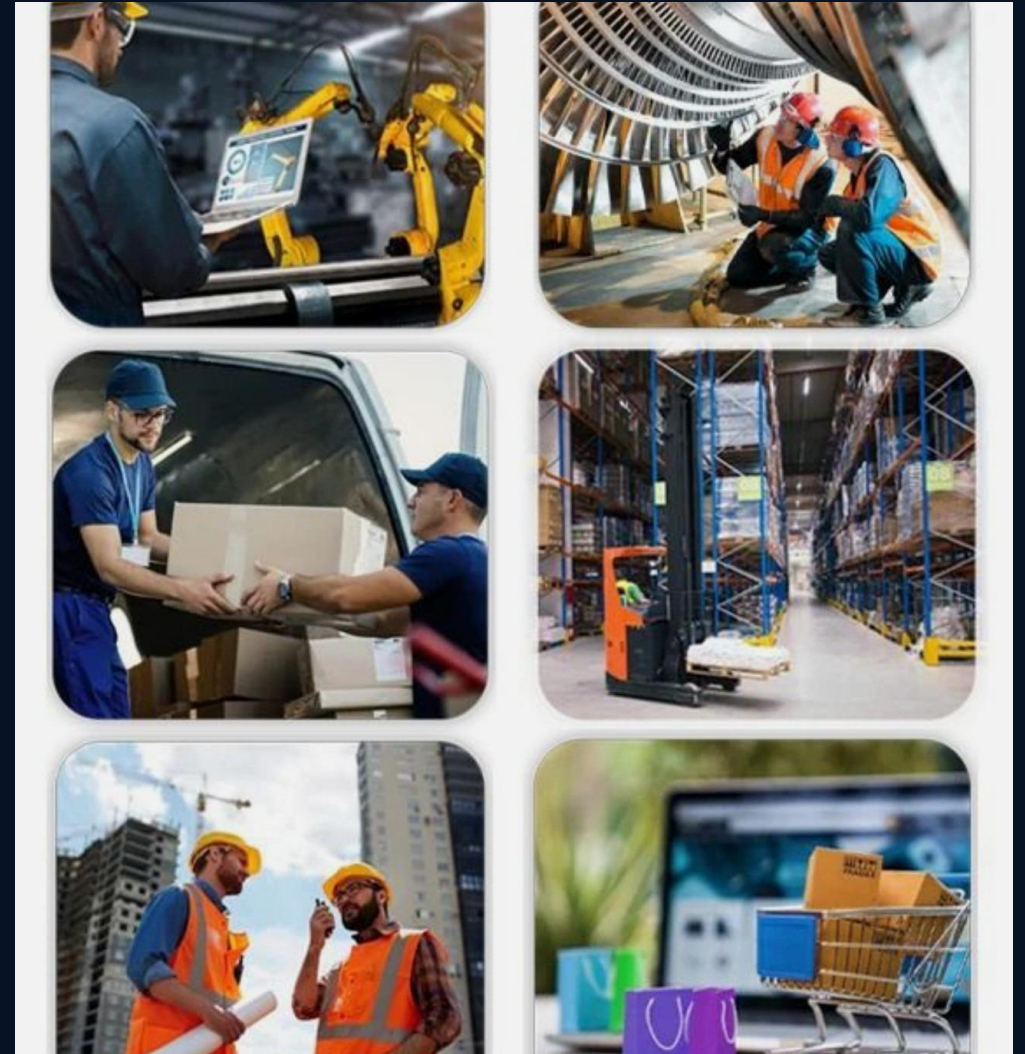
13. Key Takeaways

CONCLUSION

A concise message for the checkpoint and final demo.

- 01 See the defect** Use AI-assisted inspection and localization where data supports it.
- 02 Find the pattern** Connect defects with batch/process/production context.
- 03 Measure the impact** Identify bottlenecks and estimate throughput/economic effects.
- 04 Explain the action** Show evidence, confidence and advisory recommendations.

One system. One chain. Better factory decisions.



Appendix: Suggested Research Questions

Use these questions to guide dataset analysis once the organizer data arrives.

- 01 Which defect classes are present, and how are they labelled?
- 02 Can defects be localized with bounding boxes, masks or other annotations?
- 03 Which batch, process and production fields are available?
- 04 Which variables can be correlated with defect frequency?
- 05 How are processing times, downtime and station capacity represented?
- 06 Which economic fields are available for estimating loss or impact?
- 07 How should uncertainty and novel defects be represented?
- 08 What evidence is sufficient before making an advisory recommendation?